

2 Biological molecules

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Monomers condense into polymers, releasing water; hydrolysis breaks bonds with water. Carbohydrates have glycosidic bonds, proteins peptide bonds and nucleic acids phosphodiester bonds. The sequence of amino acids determines how a protein folds and thus its specificity; denaturation disrupts higher structure without usually hydrolysing peptide bonds.

Worked example: follow the reasoning

Two amino acids join by a condensation reaction, releasing one water molecule; a tripeptide formed from three free amino acids releases two.

Quick check · multiple choice

A What bond joins amino acids in a protein?

[1]

- A Glycosidic
- B Peptide
- C Ester
- D Hydrogen only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 How many water molecules form when one chain of seven amino acids assembles from free amino acids? Explain.

[2]

Answers and explanation

Quick check: B. Peptide

Question 1 [2 marks]

Six peptide bonds are formed [1], so six condensation events produce six water molecules [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.